

Total No. of Questions : 10]

P3309

[5353]-183

T.E. (Computer Engineering) (Semester - I)

DATA COMMUNICATION & WIRELESS SENSOR NETWORKS

(2012 Pattern)

Time : 2.30Hours]

Instructions to the candidates:

- 1) Attempt Q1 OR Q2, Q3 OR Q4 10 Marks each.
- 2) Attempt Q5 OR Q6 16 Marks each.
- 3) Attempt Q7 OR Q8 18 Marks each.
- 4) Attempt Q9 OR Q10 16 Marks each.
- 5) Neat diagrams must be drawn wherever necessary.
- 6) Assume Suitable data if necessary.
- 7) Figures to the right indicate full marks.

[Max. Marks : 70

SEAT No. :

[Total No. of Pages : 3

OR

- Q4) a) Identify the difference between Ad hoc and cellular network. [6]
- b) Write in detail applications of Wireless body Sensor Network in health domain. [4]

- Q5) a) Design suitable criteria that decides selection of schedule or contention based protocols in MAC layer. [8]
- b) Explain with neat diagram Sensor Medium Access Control Protocol in Wireless Sensor Network. [8]

OR

- Q6) a) Discuss in detail design issues related to address and name management in Wireless Sensor Network [8]
- b) Describe the necessity of low duty cycle protocols and wakeup concept in Wireless Sensor Network [8]

- Q7) a) Explain the concept of "Information via Negotiation" in SPIN. [8]
- b) Explain in detail data centric or attribute based routing protocol with example. [5]
- c) Describe role of network layer in data dissemination and gathering. [5]

OR

- Q8) a) "Low Energy Adaptive Clustering Hierarchy protocol improves lifetime of the Wireless Sensor Network" State True or False with Justification. [8]
- b) With neat diagram explain working of Power Efficient Gathering in Sensor Information System protocol in Wireless Sensor Network. [5]
- c) Compare proactive and reactive routing protocol design issues in Wireless Sensor Network [5]

- Q9) a) "Tiny OS is application specific operating system for Wireless Sensor Network" Justify the statement with respect to architecture, design, issues, functions. [8]
- b) Design role of Wireless Sensor Network in Internet of Things. [8]

[5353]-183

-2-

P.T.O

- Q1) a) "In Adaptive delta modulation quantization error increases as slope error reduces" State true or false with proper justification. [6]

- b) How does Virtual Private Network work? Write applications of VPN. [4]

OR

- Q2) a) Draw and Explain B8ZS, AMI line coding techniques in detail. [6]

- b) Explain necessity of flow and error control protocols in Wireless Sensor Network. [4]

- Q3) a) Three thousand six hundred reservation stations are available for use of single slotted ALOHA channel. The average station have 10 reservation request per hour. A slot has 125 μ s. What is approximate channel load? [6]

- b) "CSMA CD minimizes the recovery time post collision while CSMA CA reduces possibility of a collision" State true or false with justification. [4]

OR

- Q10/a) "Designing algorithms for Localization and positioning of Wireless Sensor Network in remote, disastrous region is challenging" Justify with example. [8]
- b) Discuss necessity of big data solutions for Wireless Sensor Network [4]
- c) Explain significance of anchor placement in Wireless Sensor Network [4]

